

MakaziHub

Property Management System

Technical Documentation

Version 1.0 | February 2026

Built with Laravel 11 + MySQL + Tailwind CSS

Laravel 11
PHP 8.2

MySQL 8
XAMPP

Africa's Talking
SMS API

Daraja
M-Pesa API

1. Project Overview

MakaziHub is a full-stack property management system built for Kenyan landlords and property managers. It provides end-to-end management of properties, tenants, leases, invoicing, M-Pesa payments, SMS/email notifications, and a self-service tenant portal — all from a single web-based dashboard.

1.1 Purpose & Goals

- Digitise rent collection and replace manual record-keeping
- Automate invoice generation and overdue reminders
- Integrate M-Pesa STK Push so tenants can pay from their phones
- Provide tenants with a self-service portal to view leases, invoices and payment history
- Send real-time SMS and email notifications for key events
- Generate PDF invoices and receipts automatically

1.2 User Roles

Role	Access Level	Description
SuperAdmin	Full — all modules	Platform owner, manages all properties and managers
Manager	Manager portal	Manages tenants, leases, invoices, payments and notifications
Tenant	Tenant portal only	Self-service — views lease, pays invoices, downloads receipts

1.3 Technology Stack

Layer	Technology
Backend Framework	Laravel 11 (PHP 8.2)
Database	MySQL 8 via XAMPP
Frontend Styling	Tailwind CSS (custom design tokens)
Reactive UI	Alpine.js
PDF Generation	DomPDF (barryvdh/laravel-dompdf)
SMS	Africa's Talking SMS API
Email	Laravel Mail — SMTP (Mailtrap sandbox / cPanel production)
M-Pesa	Safaricom Daraja API (STK Push + Callback)
Asset Bundling	Vite
Local Dev	XAMPP (Apache + MySQL on Windows)
Production Hosting	Truehost cPanel Shared Hosting

2. System Architecture

MakaziHub follows the standard Laravel MVC pattern with role-based routing middleware separating the three portals. All business logic lives in Service classes, keeping controllers thin.

2.1 Application Structure

```
makazihub/
├── app/
│   ├── Console/Commands/
│   │   ├── SendOverdueReminders.php
│   │   └── SendLeaseExpiryWarnings.php
│   ├── Http/Controllers/
│   │   ├── Manager/           # Manager portal controllers
│   │   ├── Tenant/           # Tenant portal controllers
│   │   └── SuperAdmin/        # SuperAdmin controllers
│   ├── Models/                # Eloquent models (10 models)
│   ├── Services/
│   │   ├── MpesaService.php   # Daraja STK Push
│   │   ├── SmsService.php     # Africa's Talking SMS
│   │   └── NotificationService.php
│   └── Mail/                  # Mailable classes (4)
├── database/migrations/       # 10 table migrations
├── resources/views/
│   ├── layouts/               # app.blade.php + tenant.blade.php
│   ├── manager/               # Manager portal views
│   ├── tenant/                # Tenant portal views
│   ├── shared/                # Shared views (notifications)
│   └── emails/                 # Branded email templates
├── routes/
│   ├── web.php                # Public + auth routes
│   ├── manager.php            # Manager portal routes
│   ├── tenant.php             # Tenant portal routes
│   └── console.php            # Scheduled commands
└── config/
    ├── africastalking.php
    └── mpesa.php
```

2.2 Role-Based Routing

```
// web.php - role-protected route groups
Route::prefix('manager')->name('manager.')
    ->middleware(['auth', 'role:manager'])
    ->group(base_path('routes/manager.php'));

Route::prefix('tenant')->name('tenant.')
    ->middleware(['auth', 'role:tenant'])
    ->group(base_path('routes/tenant.php'));
```

3. Database Schema

MakaziHub uses 10 MySQL tables. All use auto-incrementing primary keys and timestamps.

Table	Purpose
users	All accounts — name, email, password, role (superadmin/manager/tenant), phone
properties	Properties — name, address, manager_id (FK → users)
units	Rental units — unit_number, type, floor, bedrooms, monthly_rent, status (vacant/occupied)
tenants	Tenant profiles — user_id (FK), national_id, emergency_contact
leases	Lease agreements — tenant_id, unit_id, start/end dates, monthly_rent, deposit, status
invoices	Monthly invoices — invoice_number, amounts, balance, status, due_date, period dates
payments	Payment records — invoice_id, amount, method (mpesa/cash/bank), mpesa_code, paid_at
receipts	Auto-generated receipts — receipt_number, invoice_id, payment_id
notifications_log	Delivery log — user_id, type, channel (sms/email), status (sent/failed), message
password_reset_tokens	Laravel standard password reset

3.1 Key Relationships

- Property has many Units
- Unit belongs to Property, has many Leases
- Tenant has one User account, has many Leases
- Lease belongs to Tenant + Unit + Property, has many Invoices
- Invoice has many Payments, has one Receipt
- Payment belongs to Invoice
- NotificationLog belongs to User

3.2 Invoice Lifecycle

Invoices move through a defined status flow:

```
unpaid → partial (some payment made) → paid (fully settled)
unpaid → overdue (past due_date, no payment)
```

4. Development Modules

The system was built across 8 sequential modules, each adding a layer of functionality.

Module	What Was Built
1 — Database	10 migrations, seeders for all 3 roles with demo data
2 — Auth & Routing	Laravel Breeze customised, role middleware, role-based redirects after login
3 — UI System	Custom Tailwind config (Navy + Emerald), shared layouts, Alpine.js sidebar/modals
4 — Properties & Units	Full CRUD for Properties and Units, occupancy status tracking
5 — Tenants & Leases	Tenant onboarding (User + Tenant in one transaction), lease management
6 — Invoices & Payments	Invoice generation, M-Pesa STK Push, manual payment recording, PDF receipts via DomPDF
7 — Notifications	Africa's Talking SMS + Laravel Mail, NotificationService, scheduled reminders
8 — Tenant Portal	Self-service portal: lease view, invoice list/download, payment history, STK Push

4.1 Notification Events

The NotificationService dispatches both SMS and email for each trigger:

Event	Trigger	Recipients
Invoice Generated	Manager creates invoice	Tenant (SMS + Email)
Payment Received	Payment recorded	Tenant (SMS + Email)
Overdue Reminder	Daily cron — day 1, 7, 14 after due	Tenant (SMS + Email)
Lease Expiry Warning	Daily cron — 30, 14, 7 days before end	Tenant + Manager (SMS + Email)
Welcome Tenant	New tenant account created	Tenant (SMS + Email)

5. External APIs & Integrations

5.1 Safaricom Daraja API — M-Pesa STK Push

Daraja API Details

Provider: Safaricom (Kenya)

Sandbox URL: <https://sandbox.safaricom.co.ke>

Production URL: <https://api.safaricom.co.ke>

Auth: OAuth 2.0 — Consumer Key + Secret → Bearer Token

Feature: Lipa na M-Pesa Online (STK Push)

Sandbox Paybill: 174379

Sandbox Passkey:

bfb279f9aa9bdbcf158e97dd71a467cd2e0c893059b10f78e6b72ada1ed2c919

Two API calls per payment initiation:

1. POST /oauth/v1/generate?grant_type=client_credentials — get Bearer token
2. POST /mpesa/stkpush/v1/processrequest — send STK Push to phone

STK Push request payload:

```
{
  "BusinessShortCode": "174379",
  "Password": "base64(shortcode + passkey + timestamp)",
  "Timestamp": "20260227120000",
  "TransactionType": "CustomerPayBillOnline",
  "Amount": 5000,
  "PartyA": "254712345678",
  "PartyB": "174379",
  "PhoneNumber": "254712345678",
  "CallBackURL": "https://yourdomain.com/mpesa/callback",
  "AccountReference": "INV-2026-00001",
  "TransactionDesc": "Rent Payment"
}
```

The callback at /mpesa/callback auto-records the payment when Safaricom confirms:

```
// MpesaService.php - callback handler
if ($result['ResultCode'] === 0) {
    $code = collect($items)-
    >firstWhere('Name', 'MpesaReceiptNumber')['Value'];
    $amount = collect($items)->firstWhere('Name', 'Amount')['Value'];
    // Record Payment, update Invoice balance, generate Receipt
}
```

M-Pesa Environment Variables

```
MPESA_CONSUMER_KEY=your_consumer_key
MPESA_CONSUMER_SECRET=your_consumer_secret
MPESA_PAYBILL=174379
MPESA_PASSKEY=bfb279f9aa9bdbcf158e97dd71a467cd2e0c893059b10f78e6b72ada1ed2c919
MPESA_CALLBACK_URL=https://yourdomain.com/mpesa/callback
MPESA_SANDBOX=true
```

5.2 Africa's Talking — SMS API

Africa's Talking SMS

Provider: Africa's Talking

Docs: <https://developers.africastalking.com/docs/sms>

Auth: API Key in header (apiKey)

Sandbox URL: <https://api.sandbox.africastalking.com/version1/messaging>

Production URL: <https://api.africastalking.com/version1/messaging>

Note: In sandbox mode, do NOT pass a Sender ID ('from' field) — it will be rejected

Phone numbers are auto-normalised from 07XX to +254XX format before sending:

```
// SmsService.php
Http::withHeaders([
    'apiKey' => config('africastalking.api_key'),
    'Accept' => 'application/json',
])->asForm()->post($this->baseUrl, [
    'username' => config('africastalking.username'), // 'sandbox' for testing
    'to'       => $this->normalizePhone($phone),      // +254712345678
    'message'  => $message,
    // 'from' => $senderId ← OMIT in sandbox mode
]);
```

Africa's Talking Environment Variables

```
AT_API_KEY=your_api_key
AT_USERNAME=sandbox          # use 'sandbox' for testing
AT_SENDER_ID=MAKAZIHUB       # registered sender ID (production only)
AT_SANDBOX=true
```

5.3 Mailtrap — Email Testing

Mailtrap SMTP (Sandbox)

Provider: Mailtrap.io

Purpose: Catches all outgoing emails — no real emails sent during development

Host: sandbox.smtp.mailtrap.io | Port: 2525 | Encryption: TLS

Four Mailable classes were created with branded HTML email templates:

- InvoiceGeneratedMail — new invoice created
- PaymentReceivedMail — payment confirmed
- OverdueReminderMail — rent overdue reminder
- WelcomeTenantMail — new tenant onboarding

Email Environment Variables

```
MAIL_MAILER=smtp
MAIL_HOST=sandbox.smtp.mailtrap.io
MAIL_PORT=2525
MAIL_USERNAME=your_mailtrap_username
MAIL_PASSWORD=your_mailtrap_password
MAIL_ENCRYPTION=tls
MAIL_FROM_ADDRESS=noreply@makazihub.co.ke
MAIL_FROM_NAME="MakaziHub"
```

5.4 DomPDF — PDF Generation

barryvdh/laravel-dompdf

Package: barryvdh/laravel-dompdf

Install: composer require barryvdh/laravel-dompdf

Usage: Pdf::loadView('view', \$data)->setPaper('a4')->download('file.pdf')

Templates: Blade files with inline CSS for consistent PDF rendering

PDF generation is available for invoices and receipts. Both the manager and tenant can download PDFs. The same Blade templates are reused across both portals.

6. Scheduled Tasks

Two Artisan commands run automatically via a single cron job on the server:

```
# cPanel Cron Job
* * * * * php /home/username/makazihub/artisan schedule:run >> /dev/null 2>&1
```

Command	Schedule & Action
send-overdue-reminders	Daily 09:00 — sends SMS + email to tenants with invoices overdue by 1, 7, or 14 days
lease-expiry-warnings	Daily 08:00 — sends SMS + email when a lease is 30, 14, or 7 days from expiry

```
// routes/console.php
Schedule::command('send-overdue-reminders')->dailyAt('09:00');
Schedule::command('lease-expiry-warnings')->dailyAt('08:00');
```

7. Deployment Guide

MakaziHub is deployed on Truehost shared cPanel hosting. The subdomain document root points directly to the Laravel public/ folder.

7.1 Server Requirements

- PHP 8.2+ with: mbstring, openssl, pdo_mysql, tokenizer, xml, ctype, json, bcmath, fileinfo
- MySQL 5.7+ or MySQL 8
- mod_rewrite enabled (Apache)
- Composer 2.x (for local install before upload)
- Node.js 18+ (for npm run build locally)

7.2 Deployment Steps

3. Run locally: `composer install --optimize-autoloader --no-dev`
4. Run locally: `npm run build` (compiles CSS/JS to public/build/)
5. Upload project to server: `/home/username/makazihub.yourdomain.com/`
6. Set subdomain document root to:
`/home/username/makazihub.yourdomain.com/public`
7. Create .env on server with production DB and API credentials
8. Generate APP_KEY locally: `php artisan key:generate --show` — paste into .env
9. Create MySQL database + user in cPanel → MySQL Databases, grant All Privileges
10. Run: `php artisan migrate --force`
11. Run: `php artisan db:seed --force`
12. Set permissions: `chmod -R 775 storage && chmod -R 775 bootstrap/cache`
13. Clear boot cache: delete all files inside bootstrap/cache/
14. Run: `php artisan storage:link`
15. Cache for production: `php artisan config:cache && route:cache && view:cache`
16. Add cron job in cPanel for the Laravel scheduler

7.3 Production .env Checklist

```
APP_ENV=production
APP_DEBUG=false
APP_URL=https://yourdomain.com
SESSION_DRIVER=file
CACHE_STORE=file
QUEUE_CONNECTION=sync
AT_SANDBOX=false
MPESA_SANDBOX=false
MPESA_CALLBACK_URL=https://yourdomain.com/mpesa/callback
```

7.4 Common Issues

Issue	Fix
500 error on load	Check storage/logs/laravel.log — usually a missing .env value or permission issue
CSS/JS missing	Ensure npm run build ran locally and public/build/ was uploaded
Class 'files' not found	Delete all files inside bootstrap/cache/ — stale cache from local environment
Migrations fail	Check DB credentials in .env match the cPanel MySQL user you created
Storage files not accessible	Run php artisan storage:link to create the public/storage symlink

8. Security

Area	Implementation
Authentication	Laravel session-based auth, bcrypt password hashing
Authorisation	Role middleware on every route group, model-level scope checks in controllers
CSRF	@csrf on every form, Laravel auto-validates all POST/PUT/DELETE requests
SQL Injection	All queries use Eloquent ORM or parameterised query builder — no raw SQL
XSS	Blade {{ }} auto-escapes all output — {!! !!} never used for user input
Tenant Isolation	Every query is scoped to auth()->id() or the tenant's ID — no cross-access
API Keys	All keys in .env only — never committed to version control
APP_DEBUG	Set to false in production — prevents stack trace exposure
Setup Scripts	setup.php deleted immediately after use

9. Demo Accounts

Default Seeded Credentials

SuperAdmin | admin@makazihub.co.ke | Admin@1234
Manager | manager@makazihub.co.ke | Manager@1234
Tenant | tenant@makazihub.co.ke | Tenant@1234

Important: Change all passwords before going live.

10. Composer Packages

Package	Purpose
laravel/framework 11.x	Core Laravel framework
laravel/breeze	Authentication scaffold (customised)
barryvdh/laravel-dompdf	PDF generation for invoices and receipts
guzzlehttp/guzzle	HTTP client for external API calls

11. NPM Packages

Package	Purpose
tailwindcss	Utility-first CSS framework with custom design tokens
alpinejs	Lightweight reactive JS — sidebar, dropdowns, modals
vite	Fast frontend asset bundler
@tailwindcss/forms	Form element style resets

MakaziHub — Built for Kenyan Property Management

Laravel 11 · MySQL · Africa's Talking SMS · Safaricom Daraja M-Pesa · Tailwind CSS